# Pleural Mesothelioma Methylome Changes Across Histologies and in Recurrent Disease using Liquid Biopsies

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**Abstract #25-227**

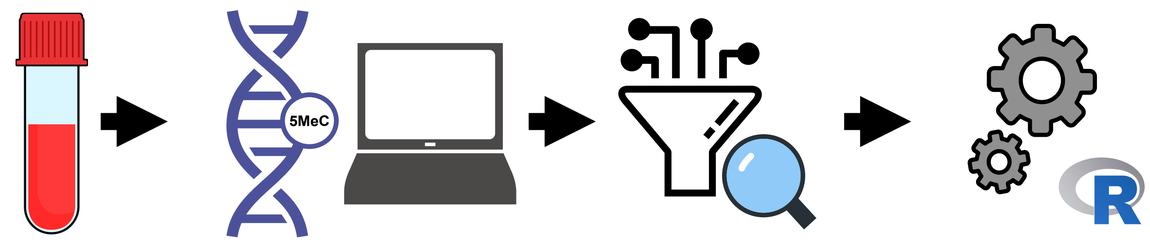
## BACKGROUND

- **Pleural mesothelioma (PM)** is complex due to molecular heterogeneity
- PM features rare recurrent mutations, making it **difficult to target** therapeutically
- Cell-free methylated DNA immunoprecipitation sequencing (**cfMeDIP-seq**) enables **non-invasive methylome profiling**

## HYPOTHESES

- cfMeDIP-seq can help identify differentially methylated regions (**DMRs**) between **PM and healthy tissues**
- cfMeDIP-seq can help **identify methylation variations** across different **PM histologies**
- cfMeDIP-seq can help **examine epigenetic changes** that occur at **disease relapse in PM**

## METHODS




Plasma from 55 PM* (chemonaive) & 24 non-cancer controls (NCCs)

*10 paired blood samples from chemonaive & recurrent PM

Libraries sequenced to depth of **70 million reads**/sample & reads binned into 300bp windows(chr. 1-22)

NCCs bins with β < 0.3 & CG-density > 2 (n = **3,537,691 windows** retained)

DMRs & pathway analysis

- **cfMeDIP-seq** performed on PM and NCC plasma samples
- DMR and pathway enrichment analyses conducted using R packages **limma**, **clusterProfiler**, **ChIPseeker**, and **annotatr**
- Lasso regression identified windows for baseline/relapse comparisons.

## RESULTS

### Cohort overview

| Characteristic | N (%) |
|---|---|
| **Sex** | |
| Female | 8 (15) |
| Male | 47 (85) |
| **Age at diagnosis (years)** | 70.5 (64.7 to 74) |
| **Histology** | |
| Biphasic | 6 (11) |
| Epithelioid | 40 (73) |
| Sarcomatoid | 9 (16) |
| **Asbestos exposure** | |
| No | 10 (18) |
| Yes | 40 (73) |
| Unknown | 5 (9) |

**Table 1.** Overview of PM cohort (N = 55) by sex, age at diagnosis, clinical stage, histology, and past asbestos exposure.

### cfMeDIP-seq identifies DMRs between PM & NCC

3,537,691 windows → T-test for each window for all PMs & NCCs → FDR correction (alpha = 0.05) → 141 windows significantly different between PM & NCC

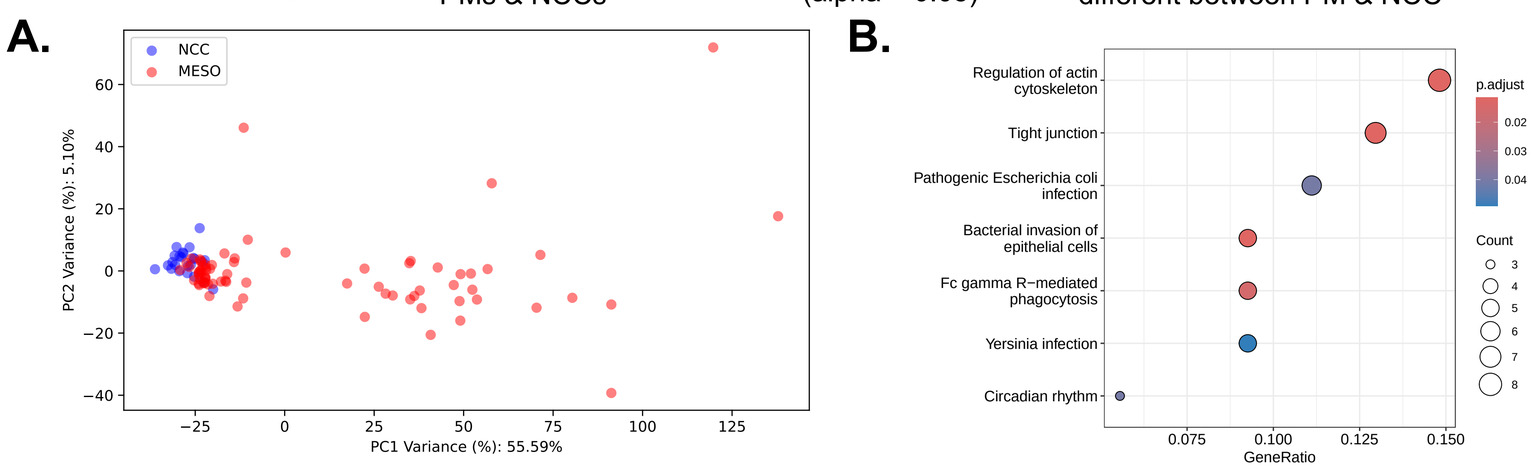



**Figure 1.** DMR analysis of PM vs. non-cancer control (NCC) identified 141 significantly different windows (adjusted p < 0.05). These windows are examined on a principal component plot (PCA) (A). KEGG pathway analysis (B) indicated significant enrichment in actin cytoskeleton and tight junction regulation genes.

### cfMeDIP-seq identifies DMRs across PM histologies

3,537,691 windows → DMR analysis between Epi. & Sarc. → FDR correction (alpha = 0.05) → 1585 windows significantly different

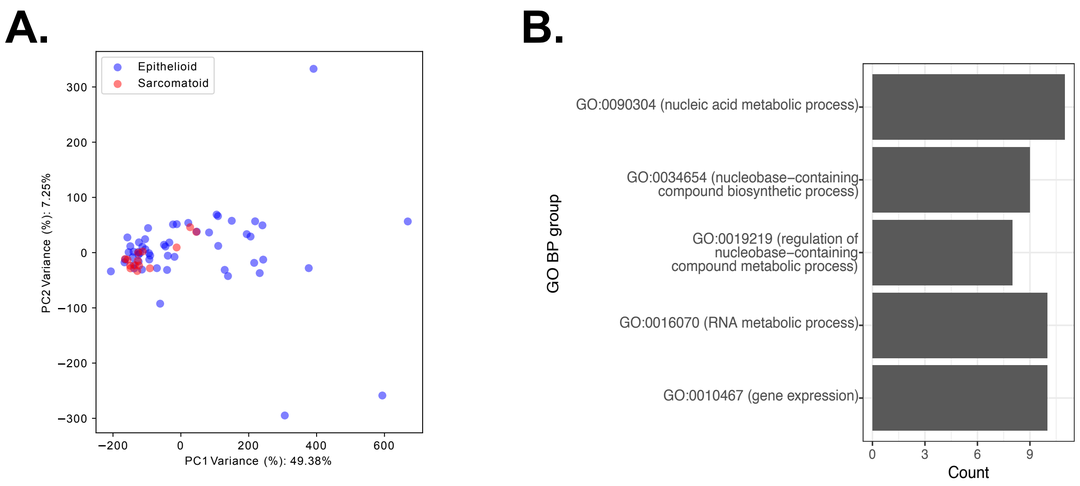



**Figure 2.** DMR analysis of epithelioid vs. sarcomatoid identified 1,585 significantly different windows (adjusted p < 0.05). These windows are examined on a heatmap (A). Gene ontology analysis (B) indicated significant enrichment in RNA processing pathways.

### cfMeDIP-seq identifies epigenetic changes at relapse

3,537,691 windows → Subset to promoter windows (n=49,675) → Lasso selection & L1 regularization → 19 promoter windows identified

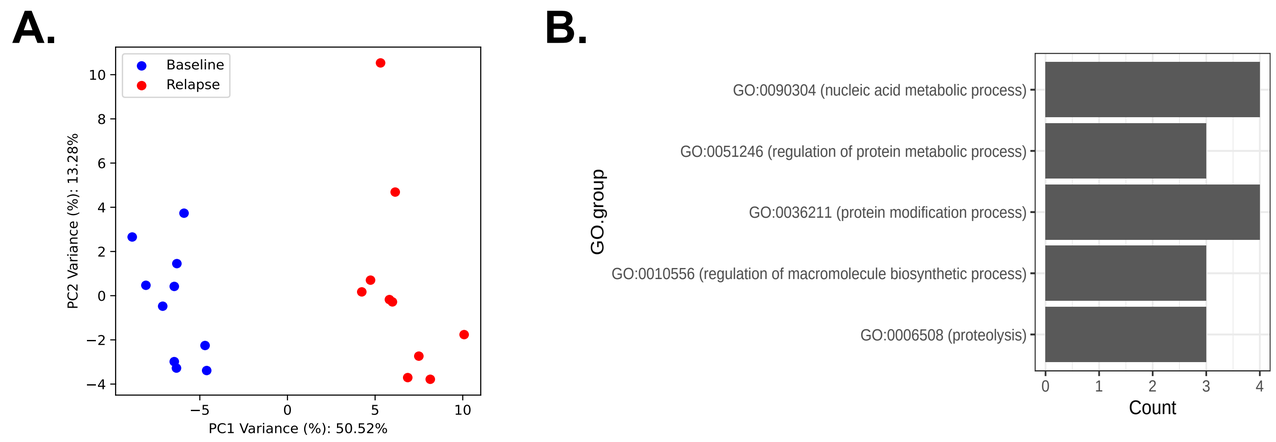



**Figure 3.** DMR analysis of 10 PM patients with blood collected at baseline and relapse. Lasso selection applied to promoter windows identified 19 windows significantly different between baseline and relapse seen in PCA (A). Gene ontology analysis (B) identified pathways enriched in nucleic acid metabolism.

## CONCLUSION

Non-invasive ctDNA profiling of PM reveals underlying biology, distinguishing cancer from non-cancer, differentiating histologies, and providing insights into changes at relapse